

Ex No: 2

IMPLEMENTATION OF DECISION MAKING AND KNOWLEDGE
REPRESENTATION

Aim:

To implement decision making and knowledge representation using prolog tool.

Prolog Code:

% Rule to find the minimum of two numbers

minimum(X, Y, X) :- X <= Y. % If X is less than or equal to Y, X is the minimum.

minimum(X, Y, Y) :- X > Y. % If X is greater than Y, Y is the minimum.

% Rule to find the maximum of two numbers maximum(X, Y, X) :- X >= Y. % If X is greater than or equal to Y, X is the maximum. maximum(X, Y, Y) :- X < Y. % If X is less than Y, Y is the maximum.

Example Queries:

1. To find the minimum of two numbers:

```
?- minimum(5, 10, Min).
```

Output:

Min = 5.

2. To find the maximum of two numbers:

```
?- maximum(5, 10, Max).
```

Output:

Max = 10.

?- minimum(8, 3, Min), maximum(8, 3, Max).

Output:

Min = 3, Max = 8.

Prolog Code:

% Given facts

likes(mary, food).

likes(mary, wine).

likes(john, wine).

likes(john, mary).

% Rules based on the conditions:

likes(john, X) :- likes(mary, X). % John likes anything that Mary likes

likes(john, Y) :- likes(Y, wine). % John likes anyone who likes wine

likes(john, Y) :- likes(Y, Y). % John likes anyone who likes themselves

% Sample queries:

% Query 1: Does John like food?

% ?- likes(john, food).

% Query 2: Does John like wine?

% ?- likes(john, wine).

% Query 3: Does John like food if Mary likes food?

% ?- likes(john, food).

% Query 4: Who does John like?

% ?- likes(john, Y).

Output:

Query: ?- likes(john, food).

yes

Query: ?- likes(john, wine).

yes

Query: ?- likes(john, food).

yes

Query: ?- likes(john, Y).

Y = mary ;

Y = john ;

Y = wine ;

Query?- likes(john, Y).

Y = mary ;

Y = john ;

Y = wine ;

Prolog Code:

```
% Rule to find the minimum of two numbers
minimum(X, Y, X) :- X <= Y. % If X is less than or equal to Y, X is the minimum.
minimum(X, Y, Y) :- X > Y. % If X is greater than Y, Y is the minimum.

% Rule to find the maximum of two numbers
maximum(X, Y, X) :- X >= Y. % If X is greater than or equal to Y, X is the maximum.
maximum(X, Y, Y) :- X < Y. % If X is less than Y, Y is the maximum.
```

Example Queries:

1. To find the minimum of two numbers:

```
?- minimum(5, 10, Min).
```

Output:

Min = 5.

2. To find the maximum of two numbers:

```
?- maximum(5, 10, Max).
```

Output:

Max = 10.

3. To find both minimum and maximum of two numbers:

```
?- minimum(8, 3, Min), maximum(8, 3, Max).
```

Output:

Min = 3, Max = 8.

Prolog Code:

```
% Given facts
likes(mary, food).
likes(mary, wine).
likes(john, wine).
likes(john, mary).
```

% Rules based on conditions:

```
likes(john, X) :- likes(mary, X). % John likes anything that Mary likes.
likes(john, Y) :- likes(Y, wine). % John likes anyone who likes wine.
likes(john, Y) :- likes(Y, Y). % John likes anyone who likes themselves.
```

% Sample queries:

4. Query 1: Does John like food?

```
?- likes(john, food).
```

5. Query 2: Does John like wine?

```
?- likes(john, wine).
```

6. Query 3: Does John like food if Mary likes food?

```
?- likes(john, food).
```

7. Query 4: Who does John like?

```
?- likes(john, Y).
```